

Recursive Offline Distillation via Epistemic Complexity Approximation

Anonymous ACL submission

Abstract

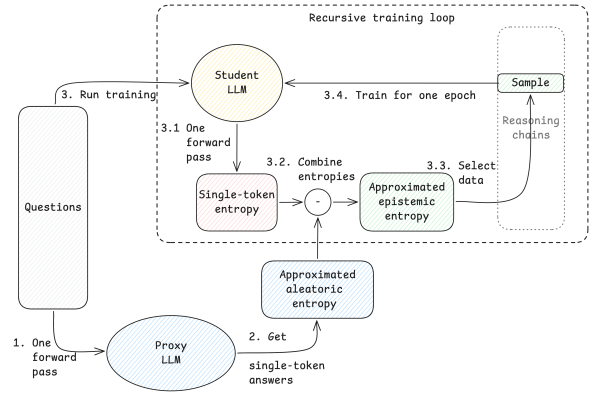


Figure 1: Recursive offline distillation scheme

Distillation of reasoning chains from Large Language Models (LLMs) into their smaller siblings is a standard training approach. Quite frequently researchers do not have full access to a teacher model and its internal representations, which limits the resulting performance. We propose a novel blueprint for efficient offline distillation that recursively uses lightweight epistemic and aleatoric complexity approximations for selective distillation. Specifically, for three open models (3-4B) at each epoch we estimate offline data complexity with a single-token answer entropy and use a medium model’s (70B) entropy as an aleatoric proxy. Next, we apply selective distillation only for a small subset of the reasoning chains with the highest corresponding epistemic component. Our approach consistently outperforms naive distillation on comparable computational budget (57.15 (± 1.00) vs 51.61 on average) and offers a much steeper learning curve (baseline final score on 15% of the data). Additionally, we present comparative analysis of methods to generate, standardize and augment synthetic reasoning chains when the teacher model does not perform perfectly on the target distillation domain in the first place. We publish our code and data to facilitate further research in this direction.

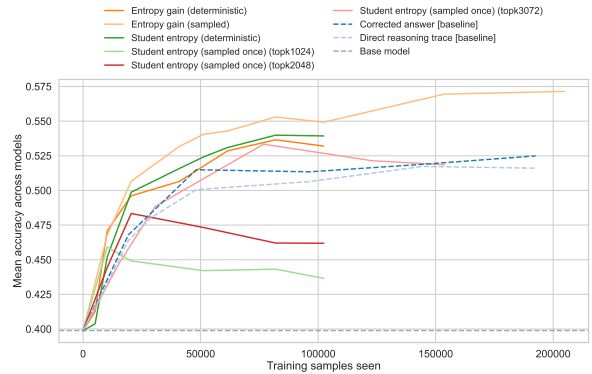


Figure 2: Mean accuracy of distillation approaches across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro

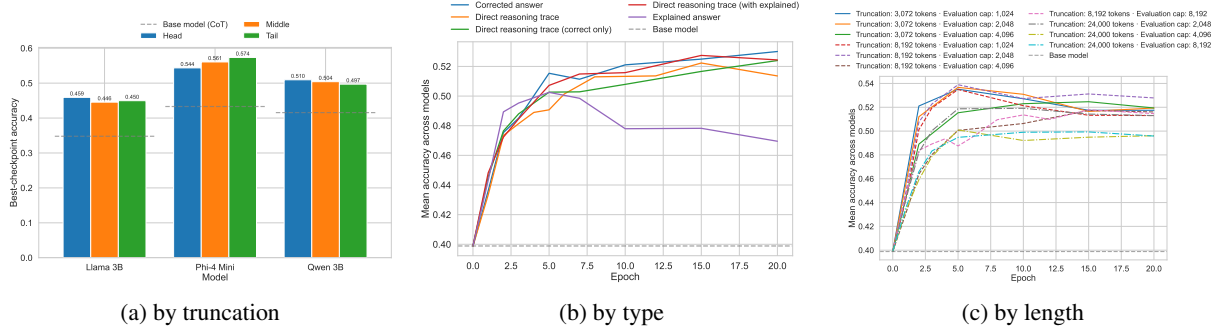


Figure 3: Comparison of synthetic traces

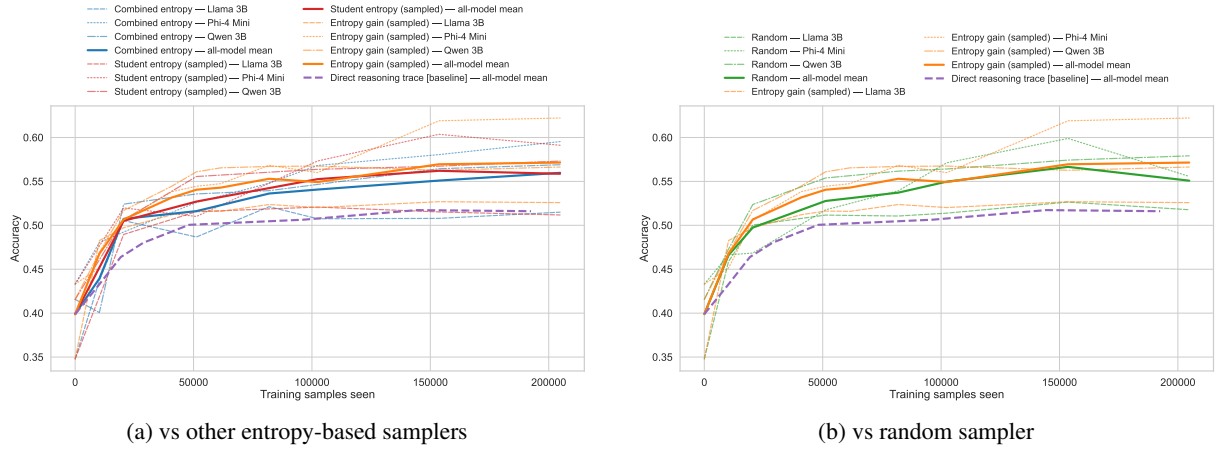


Figure 4: Comparison of entropy gain sampler

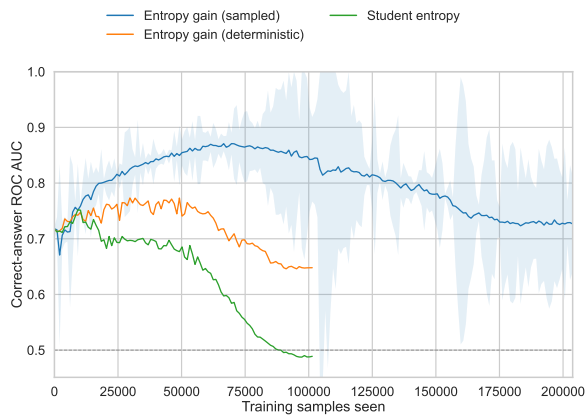


Figure 5: Single-token answer entropy dynamics

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